Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

---

1. Administrative & Study Identification

Full Study Title: A Study of Subcutaneous Nivolumab Monotherapy With or Without Recombinant Human Hyaluronidase PH20 (rHuPH20) **Short Title/Acronym:** CheckMate 8KX **Protocol Number:** CA209-8KX / BMS-986298 **NCT Number:** NCT03656718 **Posting ID:** 578228570700603510 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab subcutaneous (SC) formulation [with or without recombinant human hyaluronidase PH20 (rHuPH20)] (Trade Names: Opdivo, Opdualag | ATC Code: L01FF01) **Preferred UMLS Name:** Immunotherapy **Phase of Development:** Phase I/II (PHASE1, PHASE2) **Medical Monitor:** Clinical Trial Physician, Bristol Myers Squibb

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the pharmacokinetics (PK) of nivolumab when administered subcutaneously with or without rHuPH20. **Secondary Objectives:** To evaluate the safety profile, tolerability, and immunogenicity (anti-drug antibodies) of subcutaneous nivolumab.

2.2 Background & Rationale Nivolumab is a programmed death-1 (PD-1) inhibitor widely approved for intravenous (IV) administration across multiple cancers. Although IV nivolumab has dramatically improved clinical outcomes, it requires IV infusions that cause a significant burden on patients and healthcare systems (e.g., infusion chair time, complex scheduling, and potential port-related complications). The development of a subcutaneous (SC) formulation combined with rHuPH20 (a permeation enhancer) allows for more rapid delivery (typically < 5 minutes), which has the potential to significantly decrease the treatment burden and improve patient accessibility to the therapy.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-exploration across multiple sequential parts/ cohorts: Parts A, B, C, D, and E) **Blinding:** Open-label **Randomization:** Non-randomized (Participants allocated to specific dosage groups)

3.2 Planned Enrollment & Duration Target N: 139 participants targeted for enrollment **Number of Sites:** Multicenter / Global **Trial Status:** COMPLETED **Estimated Study Start:** 13-Jun-2018 **Estimated Primary Completion:** 07-Sep-2022 (Trial Completed)

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologic or cytologic confirmation of advanced (metastatic and/or unresectable) solid tumors of one of the following: metastatic squamous or non-squamous non-small cell lung cancer (NSCLC), advanced or metastatic renal cell carcinoma (RCC), melanoma, hepatocellular carcinoma (HCC), metastatic microsatellite instability-high or mismatch repair deficient colorectal cancer (MSI-H/dMMR CRC). In Part E, metastatic urothelial carcinoma (mUC). 2. In Part B, other solid tumor types may be considered at the discretion of the Medical Monitor. 3. Measurable disease as per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 criteria. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

4.2 Exclusion Criteria 1. Active brain metastases or leptomeningeal metastases. 2. Ocular melanoma. 3. Active, known, or suspected autoimmune disease. 4. Positive test result for hepatitis B or hepatitis C virus indicating the presence of virus (with specific protocol exceptions for patients with HCC).

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Nivolumab SC flat dose evaluated at 720 mg, 960 mg, and 1200 mg every 4 weeks (Q4W), or 600 mg every 2 weeks (Q2W), coformulated with rHuPH20 (2000 units/mL). **Route of Administration:** Subcutaneous (SC) injection manually administered by syringe into the abdomen or thigh. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or treatment discontinuation.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced malignancies that do not interfere with immune checkpoint inhibitor activity. **Prohibited:** Concurrent systemic anticancer therapy and administration of live virus vaccines.

---

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline tumor tissue availability, Laboratory screening
Baseline	Cycle 1 Day 1	First Dose of SC Nivolumab, Baseline PK/PD sampling, Safety monitoring
Interim	Every 2 or 4 Weeks	Dosing administration (Q2W or Q4W), PK/ADA blood collection, Radiographic assessment (CT/MRI) every 8 weeks for Year 1, then every 12 weeks
End of Study	Variable	Disease progression, Final PK/Safety assessment, Exit Interview

---

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Maximum Observed Serum Nivolumab Concentration (Cmax) 2. Time Taken to Reach Cmax (Tmax) 3. Area Under the Time-Serum Nivolumab Concentration Curve (AUC (TAU)), taken over the dosing interval.

Measurement Method: Blood sample analysis using a validated method for determining serum nivolumab concentrations. Collected for Arms A, B, and D on Cycle 1 Day 1, Cycle 1 Day 2, Cycle 1 Day 4, Cycle 1 Day 8, Cycle 1 Day 15, and Cycle 1 Day 21. Collected for Arm E on Cycle 1 Day 1, Cycle 1 Day 2, Cycle 1 Day 4, Cycle 1 Day 8, and Cycle 1 Day 15.

7.2 Secondary & Exploratory Endpoints * Number of Participants Experiencing Adverse Events (AEs). * Number of Participants Experiencing Treatment Related Adverse Events (TRAEs). * Number of Participants Experiencing Serious Adverse Events (SAEs). * Exploratory: Immunogenicity (incidence of anti-drug antibodies [ADA]), Pharmacodynamics (CD8+ tumor-infiltrating lymphocytes, PD-L1 expression), and patient experience/preference for SC vs. IV administration.

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Assessed continuously and graded using the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. **Serious Adverse Events (SAEs):** Collected for all participants assigned to treatment and tracked for 24-hour reporting if considered related to study drug. **Data Monitoring Committee (DMC):** Overseen by the internal Sponsor Safety Review Committee and the Clinical Trial Physician.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All treated patients (Safety Set) and all subjects with adequate samples for PK estimation (PK Set). **Sample Size Justification:** Approximately 139 participants were targeted to adequately characterize SC PK exposures; initial modeling assumed a 75% bioavailability to target exposures comparable to IV dosing. **Handling of Missing Data:** Standard approaches utilized for PK and PD endpoints; actual PK sample collection times were used instead of nominal times for parameter estimation.

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine site visits by Clinical Research Associates (CRAs) to perform Source Document Verification (SDV).

Audit Trail: Detailed tracking of eCRFs, PK sample logistics, and electronic data capture systems.

---

11. Study Identifiers & Governance

Primary ID: CA209-8KX **Registry Number**: NCT03656718 **Posting ID:** 578228570700603510 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** CheckMate 8KX **Official Regulatory Title:** Phase I/II Pharmacokinetic Multi-Tumor Study of Subcutaneous Formulation of Nivolumab Monotherapy **Disease Area:** Neoplasms by Site **MeSH Code**: D009371 **SNOMED ID**: 118742004

---

12. Clinical Framework (Template Fields Mapped to Oncology Parameters)

Primary Condition: Neoplasms by Site (including Advanced and/or metastatic solid tumors: NSCLC, RCC, Melanoma, HCC, MSI-H/dMMR CRC, mUC) **Diagnosis Threshold:** Histologic/cytologic confirmation and measurable disease per RECIST 1.1 **Medical Methodology:** Subcutaneous Immune Checkpoint Inhibitor Therapy **Optimization Target:** Target PK exposure (Cmax, Tmax, AUC) non-inferior/ comparable to historical IV administration

---

13. Study Procedures & Methodology

Management Pathway: Subcutaneous injection (abdomen or thigh) via manual syringe **Education Protocol (HCP):** Training on SC injection formulation, preparation (rHuPH20 mixing), and manual administration techniques **Education Protocol (Patient):** Education on SC vs. IV administration, tracking of patient experience and preference evaluation **Quality Audit Mechanism:** Strict PK blood sampling interval adherence and immunogenicity ADA monitoring

---

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or treatment discontinuation **Onsite Visit Cadence:** Every 4 weeks (Q4W) for Parts A-D; Every 2 weeks (Q2W) for Part E **Remote Monitoring Frequency:** As needed per standard oncology trial follow-ups **Checklist Review Interval:** Radiographic tumor assessments every 8 weeks for Year 1, then every 12 weeks

---

15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- |
| **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | |**

Clinical Operations Lead | [Name] | ___ | *[DD-MMM-YYYY]* | | *Lead Statistician | [Name]* | ____ | [DD-MMM-YYYY] |